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BioHackEU26 report for Project 10: Enabling workflows and EDAM annotations to display in community website deployments

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Introduction

Modified and extended project abstract from BioHackathon 2026:

The ELIXIR Toolkit Theme (ETT) enables diverse research and infrastructure communities to easily deploy static websites using a GitHub repository. This is beneficial as it facilitates broad and open contributions and collaborations, provides version control and development flexibility, and streamlines the process of building and populating a website. The popularity of the lightweight and standardised ETT model is demonstrated by numerous deployments, including RDMkit (Alper et al., 2025), RSQkit (Community, 2025), and WorkflowHub documentation (Gustafsson et al., 2025).

Crucially, these websites often centralise the sharing of digital resources that can be reused by specific communities. For example, the Australian Structural Biology Computing (ASBC) Community maintains an ETT website as a shared space: the intention is for the community to be able to find activities, resources, news, and events, as well as how to join the community conversation (Michie et al., 2025). In this context, *resources* refers to any digital object that supports the requirements of the community, including, but not limited to, software tools and workflows.

Including these resources in an ETT website currently requires manual curation, which is time consuming for ETT website maintainers and does not directly support making the software and workflows findable, accessible, interoperable, and reusable (FAIR) (Barker et al., 2022, wilkinson2025applying). The ideal approach would be to use the existing FAIR infrastructure ecosystem to register software and workflows, enable the ETT to interoperate with this ecosystem, and then automatically populate a website with the relevant software or workflows. This approach leverages the registries to both curate and make resources FAIR, shares the curation burden across the contributing communities, and provides the opportunity to simultaneously add ontology annotations from registry entries to ETT websites.

The ETT already offers the possibility of annotating pages with identifiers from registries like FAIRsharing (Sansone et al., 2019) and bio.tools (Mendes et al., 2026). However, support for workflows and ontologies is not implemented. This project thus worked to extend the ETT and enable creation of interactive tables to present workflows, and automation of workflow metadata import from WorkflowHub, as well as extension of the ETT integration with bio.tools to support EDAM ontology annotations at both the workflow table and ETT webpage (i.e. schema.org) level.

Ultimately, developers from any community can continue using bio.tools and WorkflowHub to make their software FAIR, and community coordinators can independently and automatically

import and embed these in an ETT deployment, benefitting from both the registry entry and its rich EDAM concept annotations. When implemented for the ETT, these features will increase the findability of the website pages and enable better integration across multiple community instances of the ETT.

Results

Project goals

The BioHackathon project had the following goals:

1. Enable interactive workflow tables for the ETT, by implementing a proof of concept integration between a community website and the WorkflowHub API.
2. Enable support for annotation with ontology terms on ETT, including import of ontology annotations from bio.tools based on existing annotations of pages.
3. Create working examples, including documentation, that can be included on the ETT test deployment page.

WorkflowHub import

Idea for table include https://australianbiocommons.github.io/workflowfinder/

Source collection on WorkflowHub: https://workflowhub.eu/collections/40

EDAM annotations

Discussion

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References

Author contributions

Federico Bianchini: Conceptualization, Supervision, Writing – original draft, Visualization; Ove Johan Ragnar Gustafsson: Conceptualization, writing – original draft.

References

Alper, P., D'Anna, F., Droesbeke, B., Andrabi, M., Buono, R. A., Bianchini, F., Bösl, K., Chandramouliswaran, I., Cook, M., Faria, D., Fatima, N., Hooft, R., Jareborg, N., Jetten, M., Pilvar, D., Poires-Oliveira, G., Popleteeva, M., Portell-Silva, L., Slifka, J., … Goble, C. (2025). RDMkit: A Research Data Management Toolkit for Life Sciences. *Patterns*. https://doi.org/10.1016/j.patter.2025.101345 **[cito:citesAsAuthority]**.

Barker, M., Chue Hong, N. P., Katz, D. S., Lamprecht, A.-L., Martinez-Ortiz, C., Psomopoulos, F., Harrow, J., Castro, L. J., Gruenpeter, M., Martinez, P. A., & Honeyman, T. (2022). Introducing the fair principles for research software. *Scientific data*, *9*(1), 622 **[cito:citesAsAuthority]**.

Community, R. (2025). *Research software quality kit (rsqkit)* (Version v1.0.0). Zenodo. https://doi.org/10.5281/zenodo.14923573 **[cito:citesAsAuthority]**.

Gustafsson, O. J. R., Wilkinson, S. R., Bacall, F., Soiland-Reyes, S., Leo, S., Pireddu, L., Owen, S., Juty, N., Fernández, J. M., Brown, T., et al. (2025). Workflowhub: A registry for computational workflows. *Scientific Data*, *12*(1), 837 **[cito:citesAsAuthority]**.

Mendes, A. I., Ienasescu, H., Andersen, C. A., Calhoun, E., Chmura, P. J., Collier, K., Goga, I., Gustafsson, O. J., Kalaš, M., Kierkegaard, M., et al. (2026). Bio. tools: An expanded web service for research software in the life sciences. *Nucleic Acids Research*, *54*(W1), W4–W12 **[cito:citesAsAuthority]**.

Michie, K. A., Litfin, T., Beecroft, S. J., Collins, B., Czabotar, P., Downton, M., Doyle, M. T., Ghosal, D., Grinter, R., Knott, G. J., Samaha, G., Christiansen, J. H., & Gustafsson, O. J. (2025). Australian structural biology deep-learning infrastructure roadmap **[cito:citesAsAuthority]**.

Sansone, S.-A., McQuilton, P., Rocca-Serra, P., Gonzalez-Beltran, A., Izzo, M., Lister, A. L., Thurston, M., & Community, F. (2019). Fairsharing as a community approach to standards, repositories and policies. *Nature biotechnology*, *37*(4), 358–367 **[cito:citesAsAuthority]**.

Appendices

If you want the Appendix (-ces) to show up after the references, wrap them in after the header, like done in this Markdown file. Look at the source to see the exact structure.